

Geometry of Splats

Chizat Bach 2018: conditional global optimization for 2-layer ReLU/Sigmoid nets via WGF, for,

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Here: $\Phi(v, W, b) = v(Wx + b)_+$, risk $R(\cdot)$, regularizer $V(\cdot)$

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Assumptions:

- R is convex, differentiable, Lipschitz on bdd sets, bdd on sublevel sets.
- V is semiconvex ($V(\theta) + \lambda |\theta|^2$ is convex for some $\lambda > 0$)
- V, Φ are 2-homogeneous functions. (satisfied by ReLU + Ridge reg.)
- μ_0 support satisfies a **separation condition**.